



COLORADO DIVISION OF MINERALS AND GEOLOGY
MINERALS PROGRAM INSPECTION REPORT
PHONE: (303) 866-3567

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The Division of Minerals and Geology has conducted an inspection of the mining operation noted below. This report documents observations concerning compliance of the mining operation with the permit and the regulations of the Mined Land Reclamation Board. The report notes 1) Areas of successful compliance; 2) Problems and suggested corrective actions and/or 3) Possible violations to be considered for possible enforcement action by the Mined Land Reclamation Board. OPERATORS SHOULD READ THIS REPORT CAREFULLY BECAUSE IT MAY REQUIRE CORRECTIVE ACTION AND/OR RESPONSES TO THE DIVISION IN ORDER TO AVOID CONSIDERATION OF POSSIBLE ENFORCEMENT ACTION BY THE MINED LAND RECLAMATION BOARD.

MINE NAME: Larimer Quarry OPERATOR: CEMEX, Inc.
COUNTY: Larimer MINERAL: SDG TYPE OF OPERATION: 112
INSPECTOR(S): Erica Crosby *[Handwritten signature: Erica Crosby]*
MINE ID # OR PROSPECTING ID #: M-1977-341
INSPECTION DATE: 9/01/04 DATE OF COMPLAINT: NA
INSPECTOR'S INITIALS: ESC TIME OF DAY (MILITARY): 10:30
INSPECTION TYPE CODE⁽¹⁾: MI POST INSP. CONTACTS⁽²⁾: NO
JOINT INSP. AGENCY CODE⁽²⁾: NO REASON FOR INSP. CODE⁽³⁾: IE
WEATHER CODE⁽⁴⁾: CR BOND CALCULATION TYPE⁽⁵⁾: BC
OP. REP. PRESENT: George Martin; CEMEX, Inc.

1. INSPECTION TYPE CODE - [CL-IN: IL=Illegal Operation, MI=Monitoring, MP=Mineral Prospect, SI=Surety-related, PR=Pre-operation]
2. POST INSPECTION CONTACTS AND JOINT INSPECTION AGENCY CODE - [CL-AG: NO=None, BL=BLM, CH=Colo. Dept. Health, CL=Land Board, CT=Citizen, CW=Wildlife, FS=Forest Service, HW=Hwy. Dept., LG=Local Government, SE=State Engr.]
3. REASON FOR INSPECTION CODE - [CL-RS: AG=Other Agency Request, CT=Citizen Complaint, IE=Normal I&E Program, HP=High Priority, PY=Priority]
4. WEATHER CODE - [CL-WE: CL=Cloudy, CR=Clear, IN=Inclement - prevented inspection, RN=Raining, SN=Snowing, WD=Windy]
5. BOND CALCULATION TYPE: - [BC=Complete Bond, BP=Partial Bond, NN=None]

This list identifies the environmental and permit parameters inspected and gives a categorical evaluation of each. IF PB OR PV IS INDICATED, YOU SHOULD READ THE FOLLOWING PAGES CAREFULLY IN ORDER TO ASSURE COMPLIANCE WITH THE TERMS OF YOUR PERMIT AND APPLICABLE RULES AND REGULATIONS. If PV is indicated, you will be notified under separate cover when the Mined Land reclamation Board will consider possible enforcement action.

GENERAL INSPECTION TOPICS

(AR) RECORDS Y (FN) FINANCIAL WARRANTY Y (RD) ROADS Y
(HB) HYDROLOGIC BALANCE Y (BG) BACKFILL & GRADING Y (EX) EXPLOSIVES Y
(PW) PROCESSING WASTE/TAILING Y (SF) PROCESSING FACILITIES Y (TS) TOPSOIL Y
(MP) GENL MINE PLAN COMPLIANCE. Y (FW) FISH & WILDLIFE Y (RV) REVEGETATION Y
(SM) SIGNS AND MARKERS Y (SP) STORM WATER MGT PLAN ... Y (SB) COMPLETE INSP. Y
(ES) OVERBURDEN/DEV. WASTE Y (SC) EROSION/SEDIMENTATION Y (RS) RECL PLAN/COMP .. Y
(AT) ACID OR TOXIC MATERIALS NA (OD) OFF-SITE DAMAGE NA (ST) STIPULATIONS NA

Y = Inspected and Found in Compliance
N = Not Inspected

PV = Inspected and Possible Violations Noted
NA = Not Applicable

PB = Inspected and Problems Noted

MINE ID # OR PROSPECTING ID # M-1977-341
INSPECTION DATE 9/01/04

INSPECTOR'S INITIALS ESC

OBSERVATIONS

This was an inspection of the Larimer Quarry conducted by Erica Crosby of the Division of Minerals and Geology. George Martin of CEMEX, Inc. was present during the inspection.

The purpose of the inspection was in regard to an acreage reduction request (AR-01) submitted by CEMEX, Inc. The request was submitted on July 12, 2004, and requested a release of 283.8 acres of the 288.8 permit area. The operator should be aware that the site is permitted for 208.8 acres (196 acres under the 1977 submittal and 12.8 acres under AM-01). Unless the operator can demonstrate that the site was permitted for 288.8 acres, the Division will use 208.8 as the permitted acreage. Regardless, the operator is requesting that all but 5 acres of the C-Pit be released.

The Larimer Quarry permit was issued on October 26, 1978 to Marten Marietta Cement for a 196-acre area. The permit was amended in 1987 to add an additional 12.8 acres to the south. On August 17, 2001, the permit was transferred to CEMEX, Inc. Technical Revision 01 (TR-01) was filed by CEMEX on August 4, 2003 to address the construction of the substation owned by Poudre Valley Rural Electric Association, Inc. and the retention of access roads throughout the site.

The quarry is divided into 4 distinct mining areas, Pits A, B, C and D. Mining and reclamation occurred concurrently until the mid-1990's. The reclamation plan calls for slopes to be reclaimed to 3h:1v, and reference is also made to 2h:1v reclaimed slopes. The required topsoil replacement depth ranges from 6"-10." The reclamation plan calls for the site to be seeded with a dryland seed mixture at a rate of 25 to 75 pounds per acre. Ammonium nitrate fertilizer is applied at a rate of 200 to 300 pounds per acre to facilitate growth. The postmining land use is rangeland habitat.

Pits A, B, portion of C and D have been reclaimed to slopes 3h:1v, replaced with topsoil and revegetated. The landowner is currently grazing the site. Grasses are well established within the permit area, and no significant erosion features were noted. The operator has achieved the postmining land use of rangeland in the release requested areas.

A five (5) acre portion of the C-Pit is excluded from the acreage reduction request. Topsoil was hauled in and spread over 5 acres of the lower area of the C-Pit in the fall of 2003. This work was conducted in response to the Division's inspection of December 6, 2002 that identified erosion problems in the area. Approximately 5,000-6,000 tons of topsoil was replaced on 4-acres of disturbed land. The area was revegetated in February, 2004 with a dryland seedmixture. Vegetation in the form of perennial grasses is establishing. A few annual weeds were observed (Blazing Star), but common for first year growth. Large erosion features were noted at the base of the slope that will need to be repaired by the operator. Gullies range in size from 1 foot to 4 feet. The establishment of vegetation should reduce the erosion potential. However, the existing gullies are large enough to require repair. The Division will not release the \$28,000.00 bond until the erosion features have been repaired and approved by the Division.

In summary, CEMEX has achieved the postmining land use in pits A, B, portions of C, and D. Therefore, the Division will approve the acreage reduction request from 208.8 acres to 5 acres (for a total reduction of 203.8 acres). The 5 acres in the C-Pit noted above and the \$28,000.00 bond will remain for Larimer Quarry.

I & E Contact Address

NAME Mr. John Lohr
OPERATOR CEMEX, Inc.
STREET P.O. Box 529
CITY/STATE/ZIP Lyons, CO 80540

- ☐ CE
- ☐ BL
- ☐ FS
- ☐ HW
- ☐ HMWMD (CH)
- ☐ SE
- ☐ WQCD (CH)

Cc: Carl Mount; DMG
Maureen Jacoby; Banks & Gesso